SIMATS ENGINEERING

ASSESSMENT TOOL 3

Annexure A – Article Review

Course Name	Artificial Intelligence
Course Code	CSA17
Course Outcome	CO4
Description	Implement machine learning techniques including decision tree algorithms, reinfo
Assessment Weightage	50%
Duration	1 hour

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Code of Conduct

I, S. Karthikeyan (Reg. No: 192419048), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ______________________________

Annexure A – Article Review

Topic area: Reinforcement Learning applications (CO4)

Task 1: Article Summary

(a) Citation, Domain and ML Problem

Y. Li, J. He, and Y. Gao, "Intelligent Traffic Signal Control with Deep Reinforcement Learning at Single Intersection," in *Proceedings of the 2021 7th International Conference on Computing and Artificial Intelligence (ICCAI '21)*, Association for Computing Machinery (ACM), New York, NY, USA, 2021, pp. 399–406. DOI: 10.1145/3467707.3467767.

Domain / Application Area: Intelligent Transportation Systems (ITS) – adaptive traffic signal control at road intersections.

Specific ML Problem Addressed: The article addresses the problem of dynamically selecting traffic signal phases and phase durations at a single intersection to minimise congestion, formulating this as a sequential decision-making problem solved via deep reinforcement learning (RL), rather than using fixed-time or simple actuated signal control.

(b) Summary (in own words)

Traditional traffic signals rely on fixed-time or rule-based actuated schedules that cannot adapt to changing traffic demand. The authors identify this as a practical gap: existing single-intersection RL controllers are usually built and tested with only one reinforcement learning algorithm, which makes it hard to know whether the reported gains come from RL in general or from the specific algorithm used. To address this, the paper implements a Proximal Policy Optimisation (PPO) based signal controller and benchmarks it directly against two other widely used deep RL approaches, Deep Q-Network (DQN) and Advantage Actor-Critic (A2C), as well as a conventional Fixed-Time controller, all within the same simulated intersection environment. By evaluating every method under identical balanced, unbalanced, saturated and unsaturated traffic conditions, the study aims to isolate the specific contribution of the RL algorithm choice to traffic signal performance, rather than simply showing that 'RL beats fixed-time control' once again.

Task 2: Methodology Analysis

(a) ML Technique and Dataset

The core technique is **Proximal Policy Optimisation (PPO)**, a policy-gradient deep reinforcement learning algorithm that learns a stochastic policy mapping traffic-state observations (e.g., queue lengths, waiting times, and signal phase) to signal-phase actions, while constraining each policy update to remain close to the previous policy for training stability. PPO is compared against the value-based **DQN** algorithm, the actor-critic **A2C** algorithm, and a non-learning **Fixed-Time** baseline.

There is no external, pre-collected dataset in the conventional sense. Instead, training and evaluation data are generated on the fly using the **Simulation of Urban MObility (SUMO)** microscopic traffic simulator, which models an isolated four-approach intersection under multiple controlled traffic-volume configurations (balanced, unbalanced, saturated, and unsaturated flows, as well as time-varying demand patterns). Each simulation episode produces a stream of state–action–reward transitions that the RL agents use for learning, so the 'dataset' is effectively the simulator's traffic-generation model rather than a fixed real-world traffic log.

(b) Mapping to CO4 and Strength/Limitation

This work maps directly onto the **Reinforcement Learning** strand of CO4: the traffic signal problem is modelled as a Markov Decision Process (state = traffic conditions, action = signal phase, reward = a function of delay/queue length), and three distinct RL algorithm families (value-based DQN, actor-critic A2C, and policy-gradient PPO) are compared as alternative solution methods.

Methodological Strength

A clear strength is the **controlled, like-for-like comparison** of three RL algorithm families and a non-RL baseline within the same simulated environment and traffic scenarios. This experimental design isolates the effect of the learning algorithm itself, giving more credible evidence than a single-algorithm study that only compares against fixed-time control.

Methodological Limitation

A key limitation is that the entire evaluation is confined to a **single, isolated intersection in simulation**. Real urban networks involve coordination across many interconnected intersections, and SUMO's driver-behaviour models are themselves simplifications of real driver psychology, so the reported advantages of PPO may not directly transfer to multi-intersection, real-world deployments without further validation.

Task 3: Results & Critical Evaluation

(a) Key Results

The paper reports simulation-based performance comparisons rather than a single classification-style metric such as accuracy or F1-score, since this is a control (not classification) task; the relevant metrics are traffic-flow measures such as average vehicle delay/waiting time and queue length across traffic scenarios. According to the authors, the **PPO-based controller outperformed DQN, A2C, and Fixed-Time control** across the tested scenarios, with the performance gain being most significant under **unbalanced traffic** (one approach saturated while others are not) and becoming only marginal when traffic was either uniformly saturated or uniformly unsaturated in all directions. PPO was additionally reported to show good portability and robustness when traffic patterns varied over time.

A qualitative comparison of the four controllers, based on the trends the authors describe, is summarised below. (Exact numerical values for delay/queue reduction are given in the paper's results tables and should be checked against the original text via the DOI above for precise figures.)

Controller	Relative Performance	Notes
Fixed-Time	Lowest	No adaptation to real-time traffic; used as baseline
DQN	Moderate	Value-based; improves over fixed-time but trails PPO
A2C	Moderate–Good	Actor-critic; competitive but less stable than PPO
PPO (proposed)	Best overall	Largest gains under unbalanced traffic; robust to time-varying demand

(b) Critical Evaluation

The reported trends are plausible and consistent with the wider RL-for-traffic literature, where policy-gradient methods like PPO are generally found to be more stable during training than value-based methods such as DQN. However, several potential sources of bias should be noted. First, all results come from a **single simulator (SUMO)** with its own idealised car-following and lane-changing models, so metrics may not generalise to real driver behaviour. Second, the study evaluates a **single isolated intersection**, which is a much simpler control problem than a coordinated network and may favour algorithms like PPO that handle a smaller state/action space well. Third, it is not stated whether results are averaged over multiple random seeds/simulation runs, which is important because deep RL training is known to have high run-to-run variance; without this, apparent performance differences could partly reflect noise rather than a genuine algorithmic advantage.

Additional experiments that would strengthen the findings include: (i) repeating training with multiple random seeds and reporting confidence intervals, (ii) extending evaluation to a multi-intersection network to test coordination and scalability, (iii) comparing against more recent specialised traffic-RL baselines (e.g., PressLight, MPLight) rather than only generic DQN/A2C, and (iv) testing under noisy or partially observed sensor data to mimic real-world detector limitations.

(c) Real-World Applicability

This technique is directly applicable to **smart-city intelligent transportation systems**, where municipal traffic-management centres could deploy learned signal-control policies at busy or highly variable-demand intersections to reduce delay and emissions. Key challenges in scaling to industry include: the **sim-to-real gap** between SUMO and actual traffic dynamics; the cost and reliability of the sensors (loop detectors, cameras) needed to provide the real-time state information the RL agent depends on; the need for a safe fallback to conventional control if the learned policy behaves unexpectedly; the substantially harder problem of **coordinating many intersections** simultaneously; and ethical/safety considerations such as ensuring pedestrian safety and emergency-vehicle priority are never compromised by an efficiency-optimising RL policy.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

- **Insight 1 – MDP formulation of control problems:** Reading this article clarified how a real-world control task (traffic signal timing) can be formally cast as a Markov Decision Process with states, actions and rewards, directly reinforcing the CO4 concept of modelling decision-making problems for reinforcement learning agents.
- **Insight 2 – Algorithm family matters, not just 'using RL':** The side-by-side comparison of DQN (value-based), A2C (actor-critic) and PPO (policy-gradient) showed that the specific RL algorithm family has a measurable effect on performance and training stability, deepening my understanding of the distinctions between value-based and policy-based RL methods covered in the syllabus.
- **Insight 3 – State/reward design is as important as the algorithm:** The fact that performance gains varied sharply with traffic scenario (large under unbalanced traffic, marginal under saturated/unsaturated traffic) shows that how a learning problem's inputs and reward signal are structured strongly shapes what the agent can learn – connecting to the broader inductive-learning principle that a model's usefulness depends heavily on how the underlying learning problem is represented, not only on the learning algorithm itself.

(b) Proposed Extension

I propose extending this single-intersection PPO controller into a **multi-agent cooperative RL framework for a small network of coordinated intersections** (e.g., a 3×3 grid), using parameter sharing across agents or a graph-attention mechanism so that each intersection's policy can condition on its neighbours' states. This would be evaluated using an open benchmark traffic-network simulator such as CityFlow or the LibSignal framework, which already provide standard multi-intersection road-network layouts and traffic demand files.

Feasibility: This extension is realistic because open-source tools (SUMO/CityFlow, LibSignal) and existing multi-agent RL algorithms already exist and are documented in the literature, so the main implementation effort is integration and hyperparameter tuning rather than building new infrastructure from scratch.

Expected Impact: If successful, this would show whether PPO's advantages at a single intersection (shown in the reviewed article) carry over to a coordinated network, where the true benefit of adaptive signal control is expected to be larger, since poor coordination between neighbouring intersections is a major real-world source of urban congestion that a single-intersection study cannot capture.